A predator-prey population model (test paper reproduce me as an ODE model.)

We consider prey X and predator Y.
The model equations are:
dX/dt = aX - bXY
dY/dt = cXY - dY
where a = 1.0, b = 0.1,
c = 0.075, d = 1.5.
Initial conditions: X(0)=10, Y(0)=5.

Discussion: the dynamics show oscillatory coexistence typical of Lotka-Volterra systems. Parameters and initial conditions are fully specified above.